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Model Predictive Control for Autonomous Vehicles: A Simulation-Based Approach

DOCTORAL THESIS

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Institute of Automation and Control
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Graz, March 2025

Declaration of the Use of AI

AI-based tools – **List tools: TOOL (LLM/Technology; Company/Developer, webpage.com)** – were used **specify how exactly the tools were used, e.g.:** in a supporting role for stylistic refinement, spelling and grammar checking, proofreading, and identifying relevant literature. All outputs were critically reviewed and, where appropriate, revised; all interpretations, analyses, and results are the author's own. **If necessary, specify further details on specific uses, if AI was used for some parts of the content of the thesis.**

Abstract

Brief outlook on the content of this document.

Kurzfassung

Kurze Beschreibung des Inhalts.

Acknowledgements

Personal acknowledgements ...

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List of Acronyms

NASA National Aeronautics and Space Administration

Chapter 1

Introduction

Content of the Introduction. The Introduction 1 holds a figure 1.1, a figure 1.2 and cites something [1]. Afterwards it continues with chapter 2. The National Aeronautics and Space Administration (NASA) was founded in 1958. The NASA is known for its space missions. Use the command $\backslash myTODO$ to mark something that needs to be done, e.g., TODO: Add more content here.

Figure 1.1: Long caption text

(a) Caption figure A

(b) Caption figure B

Figure 1.2: Overall caption

1.1 Background

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1.1.1 Sub-section 1

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1.2 Problem Statement

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List of Publications

[P1] Benedikt Andritsch, Lars Watermann, Stefan Koch, Markus Reichhartinger, Johann Reger, and Martin Horn. Modified implicit discretization of the super-twisting controller. *IEEE Transactions on Automatic Control*, 69(8):5620–5626, August 2024. ISSN 2334-3303. doi:10.1109/tac.2024.3370494.

[P2] Benedikt Andritsch, Stefan Koch, Stefan L. Hölzl, Markus Reichhartinger, and Martin Horn. Low-chattering modification of super-twisting control for insensitivity against unmodeled dynamics. *IFAC-PapersOnLine*, 59(19):79–84, 2025. ISSN 2405-8963. doi:10.1016/j.ifacol.2025.11.014.

Chapter 2

Methodology

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2.1 A table

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2.2 Some equations

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Table 2.1: Constants of a system

Parameter	Value	Unit
m	0.19	kg
k	0.2	$\frac{\mathrm{kg}}{\mathrm{s}}$
c	4.8398	$\frac{\mathrm{N}}{\mathrm{m}}$

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$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}u, \qquad y = \boldsymbol{c}^T\boldsymbol{x} + du.$$

$$\begin{bmatrix} \boldsymbol{x}_{k+1} \\ u_k \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{A}_d & \boldsymbol{\Gamma}_1(\tau_a,\tau_b) \\ \boldsymbol{0} & 0 \end{bmatrix}}_{=:\boldsymbol{A}_\xi(\tau_a,\tau_b)} \underbrace{\begin{bmatrix} \boldsymbol{x}_k \\ u_{k-1} \end{bmatrix}}_{=:\boldsymbol{\xi}_k} + \underbrace{\begin{bmatrix} \boldsymbol{\Gamma}_0(\tau_a,\tau_b) \\ 1 \end{bmatrix}}_{=:\boldsymbol{b}_\xi(\tau_a,\tau_b)} u_k, \tag{2.1}$$

$$y_k = \begin{bmatrix} \boldsymbol{c}^T & 0 \end{bmatrix} \boldsymbol{\xi}_k,$$

$$\begin{bmatrix} \boldsymbol{x}_{k+1} \\ u_k \\ u_{k-1} \\ \vdots \\ u_{k-\zeta+1} \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{A}_d & \Gamma_1(\mathrm{Y}) & \dots & \Gamma_{\zeta-1}(\mathrm{Y}) & \Gamma_\zeta(\mathrm{Y}) \\ \boldsymbol{0} & 0 & \dots & 0 & 0 \\ \boldsymbol{0} & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \boldsymbol{0} & 0 & \dots & 1 & 0 \end{bmatrix}}_{=:\boldsymbol{A}_\xi(\mathrm{Y})} \underbrace{\begin{bmatrix} \boldsymbol{x}_k \\ u_{k-1} \\ u_{k-2} \\ \vdots \\ u_{k-\zeta} \end{bmatrix}}_{=:\boldsymbol{\xi}_k}$$

$$+ \underbrace{\begin{bmatrix} \Gamma_0(\mathrm{Y}) \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{=:\boldsymbol{b}_\xi(\mathrm{Y})} u_k, \tag{2.2}$$

$$y_k = \begin{bmatrix} \boldsymbol{c}^T & 0 & 0 & \dots & 0 \end{bmatrix} \boldsymbol{\xi}_k.$$

2.3 Theorem

2.3.1 Theorem 1

Theorem 3.1: Pythagoras' theorem

In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the catheti.

$$a^2 + b^2 = c^2$$

In mathematics, the Pythagorean theorem, also known as Pythagoras' theorem, is a relation in Euclidean geometry among the three sides of a right triangle.

2.3.2 Theorem 2 with proof

Theorem 3.2

There exist two irrational numbers x, y such that x^y is rational.

Proof 3.1

If $x = y = \sqrt{2}$ is an example, then we are done; otherwise $\sqrt{2}^{\sqrt{2}}$ is irrational, in which case taking $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ gives us:

$$\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^{2} = 2.$$

References

[1] P. Peter, "Latex: A document preparation system," *Software: Practice and Experience*, vol. 23, no. 10, pp. 141–152, 1995.